Market Risk of a Commodity Inventory: VaR and Expected Shortfall Four Ways

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Abstract

A German company holds 500 tonnes of silver. Silver is priced in US dollars, the company reports in euros: the euro value of the stock moves with the silver price and with the exchange rate. We measure the one-day 95% Value at Risk (VaR) of this exposure from one year of daily data (November 2022 to November 2023), extend it to a book of silver and copper, and compare five methods: parametric (delta-normal), historical simulation, Monte Carlo with normal and Student-t shocks, and filtered historical simulation with EWMA volatility. The VaR of the silver stock (EUR 334 m) is EUR 8.3 m, i.e. 2.5 % of its value; ignoring the correlation between silver and the dollar would overstate it by 18 %, because the two act as a natural hedge in euros. For the two-commodity book the one-day estimates of the methods differ by at most 18 % and differ most in the tail (99% ES, ten days). All methods pass an out-of-sample backtest, but with half a year of backtest the tests have little power: a 99% model with twice the nominal exception rate is rejected only one time in eight.

1 The exposure

Let q_i be the quantity of commodity i (in the unit of its price), S_i its price in USD and x the EUR/USD rate (USD per EUR). The euro value of the book is

$$V = \sum_i V_i, \qquad V_i = \frac{q_i S_i}{x}. \tag{1}$$

With the risk factors $s_i = \log S_i$ and $f = \log x$, a scenario of factor changes $(\Delta s, \Delta f)$ revalues the book exactly (*full revaluation*) as

$$\Delta V = \sum_i V_i \left(e^{\Delta s_i - \Delta f} - 1 \right) \approx \sum_i V_i (\Delta s_i - \Delta f), \tag{2}$$

the right-hand side being the delta approximation. A rise of the dollar ($\Delta f < 0$) raises the euro value; when silver and the euro rise together, the two effects cancel. The loss is $L = -\Delta V$; for a level α, VaR_α is the α-quantile of L and the expected shortfall $\mathrm{ES}_\alpha = \mathbb{E}[L \mid L \geq \mathrm{VaR}_\alpha]$ the mean loss beyond it (McNeil et al., 2015). ES is a coherent risk measure (Artzner et al., 1999); VaR is not, but remains the reference in reporting.

The case. The course exercise gives 500 t of silver (16.1×10^6 troy ounces) and asks for the one-day VaR(95%) of its euro value, "taking into account the correlation that follows from the data". As a second position we add 40 000 t of copper; the course gives no quantity, so it is chosen to make the two positions of similar size (EUR 334 m and EUR 300 m on 10 November 2023).

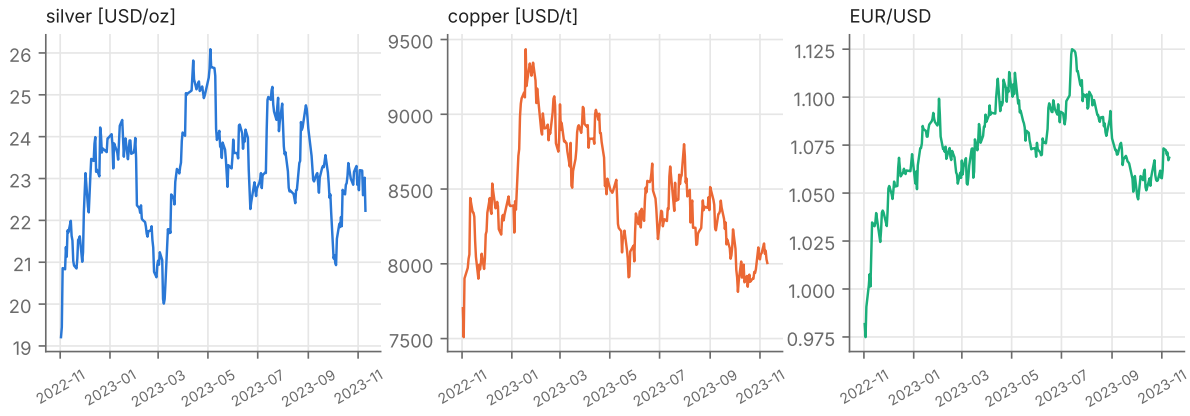


Figure 1: Daily closes used in the study (weekdays common to the three series).

Data. Daily closes of silver (USD/oz), copper (USD/t) and EUR/USD from 2 November 2022 to 10 November 2023, as provided in the course (source: finanzen.net). Silver and EUR/USD also contain weekend quotes; on the 258 weekdays on which all three closed we have 257 daily returns (Figure 1). Annualised volatilities are 27 % (silver), 20 % (copper) and 9 % (EUR/USD); silver correlates with EUR/USD at +0.48 and with copper at +0.21. All three return series have excess kurtosis (2.2, 1.4 and 0.7): fat tails, one of the stylised facts of financial returns. The course data are not redistributed; the repository contains a script that builds the same table from public sources (Chavarría Morales, 2026).

2 Methods

Parametric (delta-normal). Factor changes normal with the sample covariance Σ, delta approximation: $L \sim N(0, \delta^\top \Sigma \delta)$ with $\delta = (V_1, \ldots, V_n, -V)$, so $\mathrm{VaR}_\alpha = z_\alpha \sigma_L$ and $\mathrm{ES}_\alpha = \sigma_L \varphi(z_\alpha)/(1-\alpha)$.

Historical simulation. Today's book revalued under every observed (overlapping) h-day factor change; no distributional assumption, but only as many scenarios as days.

Monte Carlo. 200 000 correlated scenarios (Cholesky factor of Σ) with full revaluation, with normal shocks or multivariate Student-t shocks of the same covariance; the degrees of freedom ($\nu = 8.2$) are fitted to the daily P&L of the book.

Filtered historical simulation. Each factor's returns are divided by their EWMA volatility ($\lambda = 0.94$, J.P. Morgan/Reuters, 1996) and rescaled by today's forecast (Hull and White, 1998): the empirical joint shape of the shocks is kept, volatility clustering is taken into account.

For ten days, the parametric and Monte Carlo methods scale the covariance by 10 (square-root-of-time); the historical methods use overlapping ten-day returns. Risk contributions follow the Euler allocation of the parametric VaR, $C_i = z_\alpha \, \mathrm{cov}(L_i, L)/\sigma_L$, which adds up to the VaR of the book (Tasche, 1999).

3 The course exercise: 500 t of silver

The one-day VaR(95%) of the silver stock is **EUR 8.28 m**, 2.5 % of its value of EUR 334 m; the ES(95%) is EUR 10.4 m. The exercise asked for the correlation, and it matters: with the same volatilities and zero correlation, the VaR would be EUR 9.76 m, 18 % higher. Silver and the euro

Table 1: VaR and ES of the silver and copper book (EUR m), as of 10 November 2023.

	95%				99%			
	1 day		10 days		1 day		10 days	
method	VaR	ES	VaR	ES	VaR	ES	VaR	ES
parametric (delta-normal)	10.4	13.1	33.0	41.4	14.7	16.9	46.6	53.4
historical simulation	9.8	12.4	30.2	35.0	14.7	16.2	39.4	41.2
Monte Carlo, normal	10.3	12.9	31.8	39.7	14.6	16.7	44.7	51.0
Monte Carlo, Student t	10.1	13.6	31.1	41.6	15.7	19.4	47.9	58.8
filtered HS (EWMA)	8.8	13.1	27.4	39.7	14.7	19.0	45.4	57.4

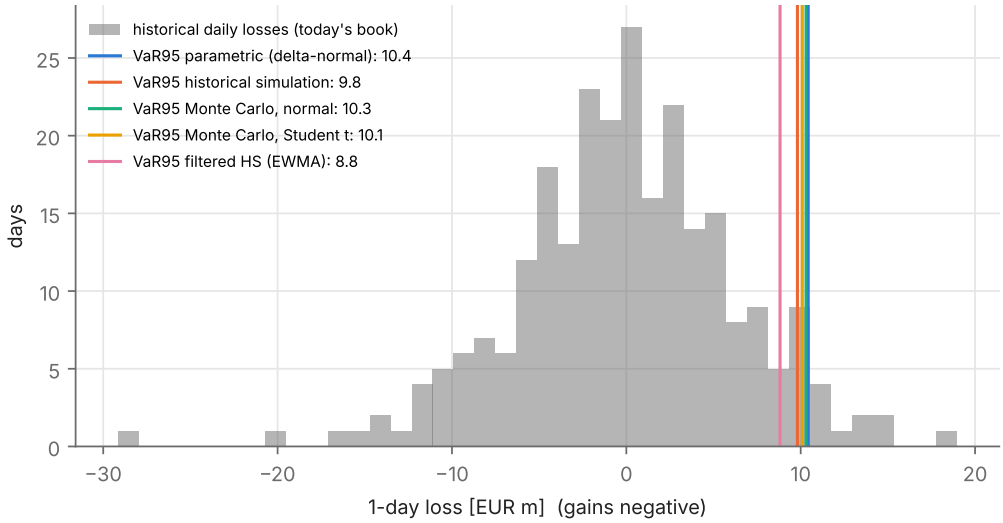


Figure 2: One-day losses of today's book under the 257 historical scenarios, and the one-day VaR(95%) of the five methods.

tend to rise together against the dollar ($\rho = +0.48$), so part of a fall of silver in USD is offset in EUR by a weaker euro and vice versa: holding silver in a euro balance sheet is less risky than its dollar volatility suggests (silver price risk alone: EUR 9.29 m).

4 The book: silver and copper

Table 1 compares the methods. At the one-day horizon and 95%, they differ by at most 18 % (EUR 8.8–10.4 m); the filtered historical simulation is lowest because volatility at the end of the sample was below its one-year average. At 99% and ten days the differences grow: the Student-t Monte Carlo and the filtered historical simulation have the largest ES (EUR 57–59 m), the plain historical simulation the smallest, because 248 overlapping ten-day returns contain few independent extreme scenarios. Figure 2 shows the historical loss distribution of today's book with the five one-day VaR estimates.

Diversification. The stand-alone VaRs of silver and copper add up to EUR 14.2 m; the VaR of the book is EUR 10.4 m, 27 % less (Table 2 and Figure 3). Silver contributes two thirds of the book VaR for a little over half of its value: copper is less volatile and only weakly correlated with silver.

Table 3: Out-of-sample backtest of the one-day VaR (rolling window of 125 days, 132 test days).

method	level	expected	exceptions	Kupiec p	Christoffersen p
parametric (delta-normal)	95%	6.6	5	0.51	0.62
historical simulation	95%	6.6	7	0.87	0.45
Monte Carlo, normal	95%	6.6	5	0.51	0.62
filtered HS (EWMA)	95%	6.6	7	0.87	0.45
parametric (delta-normal)	99%	1.3	2	0.58	0.80
historical simulation	99%	1.3	3	0.21	0.76
Monte Carlo, normal	99%	1.3	2	0.58	0.80
filtered HS (EWMA)	99%	1.3	3	0.21	0.76

Table 2: Euler allocation of the one-day parametric VaR(95%) (EUR m).

position	value [EUR m]	stand-alone VaR	contribution
silver	334	8.28	6.82
copper	300	5.92	3.61
book	634	14.20	10.43

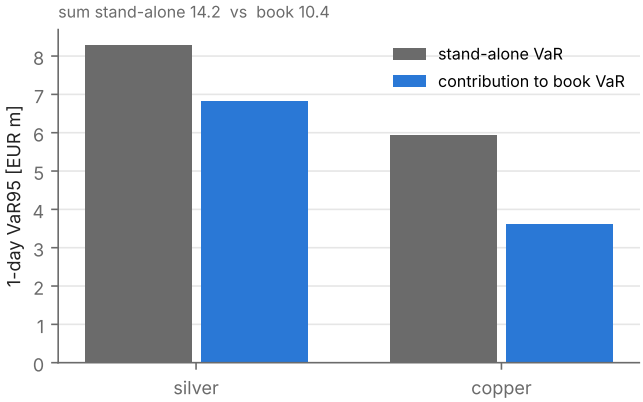


Figure 3: Stand-alone VaR and contribution to the book VaR of each position.

5 Backtest

Each day from May 2023, the one-day VaR is estimated from the previous 125 days and compared with the realised loss of the book on the next day (132 days, Figure 4). The Kupiec test checks the number of exceptions (Kupiec, 1995), the Christoffersen test their independence (Christoffersen, 1998). All methods pass both at 95% and 99% (Table 3).

A passed backtest of this length is weak evidence. Table 4 gives the probability that the Kupiec test rejects a model whose true exception rate is too high: a 95% VaR that is exceeded on 10 % of the days is caught in about half of the cases, a 99% VaR exceeded on 2 % of the days in one case out of eight. Regulatory backtests use 250 days or more for this reason, and at the 99% level even a year leaves little power.

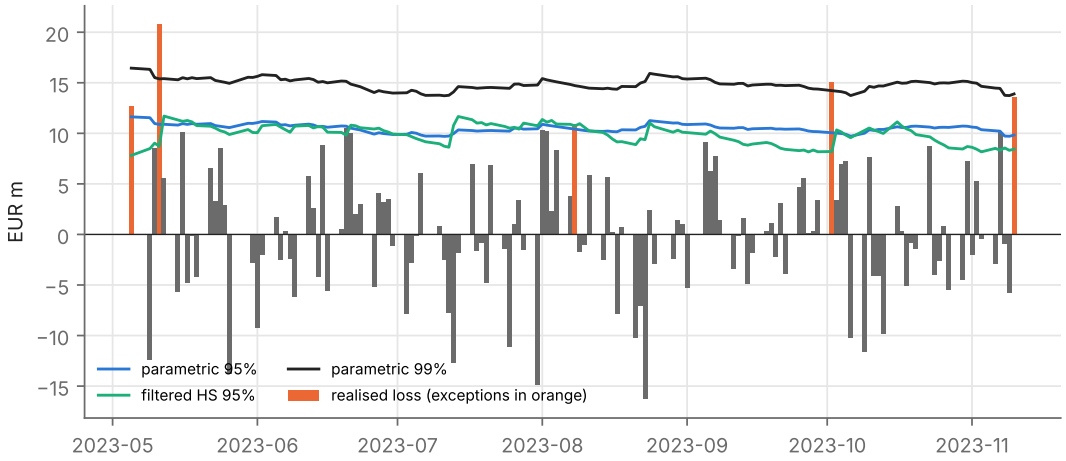


Figure 4: Realised daily losses of the book (exceptions of the parametric 95% VaR in orange) and the rolling VaR forecasts.

Table 4: Power of the Kupiec test with 132 backtest days (5% significance).

VaR level	true exception rate	probability of rejection
95%	7.5%	19%
95%	10.0%	57%
95%	15.0%	97%
99%	1.5%	5%
99%	2.0%	13%
99%	3.0%	36%

6 Main changes with respect to the course notebook

The notebook written for the exercise did not yet answer it. Table 5 lists the corrections.

Table 5: Main changes with respect to the original notebook.

	course notebook	this version
Alignment	returns of silver and EUR/USD paired by row position, not by date (correlation −0.13)	series aligned by date (correlation +0.48)
Correlated shocks	Cholesky factor applied to draws already scaled by mean and volatility, volatility applied a second time when revaluing	Cholesky factor of the covariance applied to standard shocks
Risk measure	95th percentile of each factor's return separately; one simulated day revalued	quantile and tail mean of the loss of the position
Units	500 multiplied by a price per troy ounce	$500\,\mathrm{t} = 16.1 \times 10^6$ troy ounces
Methods	Monte Carlo only	five methods, two levels, two horizons, ES
Validation	none	backtest with Kupiec and Christoffersen tests, power analysis; tests on synthetic data

7 Conclusions

1. The one-day VaR(95%) of 500 t of silver held in a euro balance sheet was EUR 8.3 m in November 2023, 2.5 % of its value. The positive correlation of silver with the euro is a natural hedge: without it the VaR would be 18 % higher.

2. For a book of silver and copper, the five methods give one-day 95% VaRs between EUR 8.8 and 10.4 m. Model choice matters in the tail: for the ten-day 99% ES the estimates range from EUR 41 to 59 m, and the methods that capture fat tails or changing volatility give the higher values.

3. Diversification between the two commodities lowers the VaR by 27 %.

4. All methods pass the backtest, but half a year of backtest cannot tell good from mediocre models at the 99% level. With the public data script, the same analysis runs on any period, and a longer history is the first improvement to make.

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